--RFM Analysis | Customer Segmentation

--data ile tanis olaq(20 sutun olduguna gore sekillerin sayi cox oldu)

SELECT * **FROM** RETAIL_SALES ;

	ROW_ID	ORDER_ID	ORDER_DATE	SHIP_DATE	SHIP_MODE	CUSTOMER_ID	CUSTOMER_NAME	SEGMENT	
1	1	CA-2016-152156	2016-11-08 00:00:00.000	2016-11-11 00:00:00.000	Second Class	CG-12520	Claire Gute	Consumer	Un
2	2	CA-2016-152156	2016-11-08 00:00:00.000	2016-11-11 00:00:00.000	Second Class	CG-12520	Claire Gute	Consumer	Un
3	3	CA-2016-138688	2016-06-12 00:00:00.000	2016-06-16 00:00:00.000	Second Class	DV-13045	Darrin Van Huff	Corporate	Un
4	4	US-2015-108966	2015-10-11 00:00:00.000	2015-10-18 00:00:00.000	Standard Class	SO-20335	Sean O'Donnell	Consumer	Un
5	5	US-2015-108966	2015-10-11 00:00:00.000	2015-10-18 00:00:00.000	Standard Class	SO-20335	Sean O'Donnell	Consumer	Un
6	6	CA-2014-115812	2014-06-09 00:00:00.000	2014-06-14 00:00:00.000	Standard Class	BH-11710	Brosina Hoffman	Consumer	Un
7	7	CA-2014-115812	2014-06-09 00:00:00.000	2014-06-14 00:00:00.000	Standard Class	BH-11710	Brosina Hoffman	Consumer	Un
8	8	CA-2014-115812	2014-06-09 00:00:00.000	2014-06-14 00:00:00.000	Standard Class	BH-11710	Brosina Hoffman	Consumer	Un
9	9	CA-2014-115812	2014-06-09 00:00:00.000	2014-06-14 00:00:00.000	Standard Class	BH-11710	Brosina Hoffman	Consumer	Un
10	10	CA-2014-115812	2014-06-09 00:00:00.000	2014-06-14 00:00:00.000	Standard Class	BH-11710	Brosina Hoffman	Consumer	Un

	COUNTRY	CITY	STATE	REGION	PRODUCT_ID	CATEGORY	SUB_CATEGORY	PRODUCT_NAME	SALES
1	United States	Henderson	Kentucky	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection E	261.96
2	United States	Henderson	Kentucky	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholst	731.94
3	United States	Los Angeles	California	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Lab	14.62
4	United States	Fort Lauderdale	Florida	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slir	957.5775
5	United States	Fort Lauderdale	Florida	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart Sys	22.368
6	United States	Los Angeles	California	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood a	48.86
7	United States	Los Angeles	California	West	OFF-AR-10002833	Office Supplies	Art	Newell 322	7.28
8	United States	Los Angeles	California	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP p	907.152
9	United States	Los Angeles	California	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders wi	18.504
10	United States	Los Angeles	California	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outle	114.9

SALES	QUANTITY	DISCOUNT	PROFIT
261.96	2	0	41.9136
731.94	3	0	219.582
14.62	2	0	6.8714
957.5775	5	0.45	-383.031
22.368	2	0.2	2.5164
48.86	7	0	14.1694
7.28	4	0	1.9656
907.152	6	0.2	90.7152
18.504	3	0.2	5.7825
114.9	5	0	34.47

--birinci cedvelin baslangic ve son tarixine baxaq

```
SELECT
        MIN(order_date) min_date,
        MAX(order_date) max_date
FROM RETAIL_SALES
```

MIN_DATE	MAX_DATE
2014-01-03 00:00:00.000	2017-12-30 00:00:00.000

--cedvel kohne tarixleri ehate etdiyine gore cedvelin son tarixini max_date kimi gostermek lazimdir

```
WITH max_date_table AS (
        SELECT
                max(order_date) AS  max_date
        FROM RETAIL_SALES
),

--indi RFM hesablayaq
RFM AS(
        SELECT
                customer_id,
                max_date - MAX(order_date) AS recency,
                COUNT(DISTINCT(order_id)) AS frequency,
                SUM(sales) AS monetary
        FROM RETAIL_SALES r
        CROSS JOIN max_date m
GROUP BY r.customer_id, m.max_date
```

),

	CUSTOMER_ID	RECENCY	FREQUENCY	MONETARY
1	DB-13060	16	7	4,531.646
2	SC-20050	40	9	4,624.572
3	TB-21355	146	5	834.328
4	AA-10315	184	5	5,563.56
5	LH-16750	151	3	287.426
6	BF-11170	16	11	5,907.972
7	AA-10480	259	4	1,790.512
8	AD-10180	29	8	6,106.88
9	ML-17395	69	9	2,722.843
10	AC-10420	232	3	925.8

```
--rfm ucun 1-5 qiymetlerinde skor verek
RFM_score AS(
        SELECT
                customer_id,
                recency,
                frequency,
                monetary,
                NTILE(5) OVER(ORDER BY recency DESC)   AS r_score,
                NTILE(5) OVER(ORDER BY frequency)              AS f_score,
                NTILE(5) over(ORDER BY monetary)               AS m_score
        FROM RFM
)
```

	CUSTOMER_ID	RECENCY	FREQUENCY	MONETARY	R_SCORE	F_SCORE	M_SCORE
1	NB-18580	1,165	2	273.872	1	1	1
2	GR-14560	1,135	2	1,284.38	1	1	2
3	RE-19405	1,097	1	48.36	1	1	1
4	CM-12715	1,035	4	3,984.4524	1	1	4
5	VT-21700	1,000	2	1,736.596	1	1	2
6	PC-19000	882	2	1,061.488	1	1	2
7	AG-10525	845	4	1,455.038	1	1	2
8	PF-19120	835	4	9,062.864	1	1	5
9	DP-13165	812	2	1,058.616	1	1	2
10	CC-12685	799	4	2,854.115	1	2	4

--skor araliqlarini yoxlayaq . burden sonraki hisse range neticesini almaq ucundur. Novbeti addimda qaldigimiz yerden davam edeceyik.

```
recency_range AS(
    SELECT
        r_score,
        MIN(recency)    AS rmin,
        MAX(recency)    AS rmax
    FROM rfm_score
    GROUP BY r_score
),
--her biri ucun ayrica hesablanmalidir
frequency_range AS(
    SELECT
        f_score,
        MIN(frequency)    AS fmin,
        MAX(frequency)  AS fmax
    FROM rfm_score
    GROUP BY f_score
),
```

```
monetary_range as(
	SELECT
		m_score,
		MIN(monetary)  AS mmin,
		max(monetary)  AS mmax
	FROM rfm_score
	GROUP BY m_score
)
SELECT
	r.r_score,
	r.rmin,
	r.rmax,
	f.f_score,
	f.fmin,
	f.fmax,
	m.m_score,
	m.mmin,
	m.mmax
FROM recency_range r
JOIN frequency_range f
	ON r.r_score=f.f_score
JOIN monetary_range m
	ON r.r_score=m.m_score
ORDER BY r.r_score
```

	R_SCORE	RMIN	RMAX	F_SCORE	FMIN	FMAX	M_SCORE	MMIN	MMAX
1	1	225	1,165	1	1	4	1	4.833	959.477
2	2	103	224	2	4	5	2	964.636	1,739.689
3	3	53	102	3	5	7	3	1,744.7	2,754.216
4	4	26	52	4	7	9	4	2,767.219	4,282.94
5	5	0	26	5	9	17	5	4,299.161	25,043.05

--RFM ortalamasini ve concat neticesini gormek

avg_rfm_concat **AS**(

```
    SELECT
        customer_id,
        recency,
        frequency,
        monetary,
        r_score ||'-'|| f_score ||'-'|| m_score AS R_F_M,
        ROUND((r_score + f_score + m_score) / 3, 2) AS AVG_rfm
    FROM rfm_score
)
```

	CUSTOMER_ID	RECENCY	FREQUENCY	MONETARY	R_F_M	AVG_RFM
1	NB-18580	1,165	2	273.872	1-1-1	1
2	GR-14560	1,135	2	1,284.38	1-1-2	1.33
3	RE-19405	1,097	1	48.36	1-1-1	1
4	CM-12715	1,035	4	3,984.4524	1-1-4	2
5	VT-21700	1,000	2	1,736.596	1-1-2	1.33
6	PC-19000	882	2	1,061.488	1-1-2	1.33
7	AG-10525	845	4	1,455.038	1-1-2	1.33
8	PF-19120	835	4	9,062.864	1-1-5	2.33
9	DP-13165	812	2	1,058.616	1-1-2	1.33
10	CC-12685	799	4	2,854.115	1-2-4	2.33

--Müştəri dəyərinə(monetary) görə Low / Mid / High Value yarat. (burada m_score’dan istifade etmek ucun evvelki layere m_score leave etdim)

```
avg_rfm_concat AS(
    SELECT
        customer_id,
        recency,
        frequency,
        monetary,
        m_score,
        r_score ||'-'|| f_score ||'-'|| m_score AS R_F_M,
        ROUND((r_score + f_score + m_score) / 3, 2) AS AVG_rfm
    FROM rfm_score
),
```

```
value_seg AS(
    SELECT
        customer_id,
        recency,
        frequency,
        monetary,
        R_F_M,
        AVG_rfm,
        CASE
            WHEN m_score in(1,2) THEN 'low'
            WHEN m_score = 3    THEN 'mid'
            WHEN m_score in(4,5) THEN 'high'
        END AS value_segment
    FROM avg_rfm_concat
)
```

	CUSTOMER_ID	RECENCY	FREQUENCY	MONETARY	R_F_M	AVG_RFM	VALUE_SEGMENT
1	NB-18580	1,165	2	273.872	1-1-1	1	low
2	GR-14560	1,135	2	1,284.38	1-1-2	1.33	low
3	RE-19405	1,097	1	48.36	1-1-1	1	low
4	CM-12715	1,035	4	3,984.4524	1-1-4	2	high
5	VT-21700	1,000	2	1,736.596	1-1-2	1.33	low
6	PC-19000	882	2	1,061.488	1-1-2	1.33	low
7	AG-10525	845	4	1,455.038	1-1-2	1.33	low
8	PF-19120	835	4	9,062.864	1-1-5	2.33	high
9	DP-13165	812	2	1,058.616	1-1-2	1.33	low
10	CC-12685	799	4	2,854.115	1-2-4	2.33	high

--davranis esasinda segmentler yaratmaq.

--burda bele error aldim ve orgendimki herhansisa sutundan istifade etmek ucun ondan bir evvelki layerde mutleq o sutundan istifade etmelisen.

```
SQL Error [904] [42000]: ORA-00904: "F_SCORE": invalid
identifier
```

--ve mende evvelki sutunlara elave etdim

```
value_seg AS(
    SELECT
        customer_id,
        recency,
        frequency,
        monetary,
        R_F_M,
        AVG_rfm,
        r_score,
        f_score,
        m_score,
        CASE
            WHEN m_score in(1,2) THEN 'low'
            WHEN m_score = 3      THEN 'mid'
            WHEN m_score in(4,5) THEN 'high'
        END AS value_segment
    FROM avg_rfm_concat
),
```

--davam edek.

customer_seg **as**(

SELECT

customer_id,

```
            recency,
            frequency,
            monetary,
            R_F_M,
            AVG_RFM,
            value_segment,
            CASE
                WHEN r_score >=4 AND f_score >=4 AND m_score >=4 THEN 'VIP'
                WHEN f_score >=3 AND m_score <4 THEN 'Regular'
                WHEN r_score <=3 AND r_score>1 THEN 'Dormat'
                WHEN r_score =1 THEN 'Churned'
                WHEN r_score >=4 AND f_score <=4 THEN 'New Customer'
            END AS cust_seg
        FROM value_seg
)
```

	CUSTOMER_ID	RECENCY	FREQUENCY	MONETARY	R_F_M	AVG_RFM	VALUE_SEGMENT	CUST_SEG
2	GR-14560	1,135	2	1,284.38	1-1-2	1.33	low	Churned
3	RE-19405	1,097	1	48.36	1-1-1	1	low	Churned
4	CM-12715	1,035	4	3,984.4524	1-1-4	2	high	Churned
5	VT-21700	1,000	2	1,736.596	1-1-2	1.33	low	Churned
6	PC-19000	882	2	1,061.488	1-1-2	1.33	low	Churned
7	AG-10525	845	4	1,455.038	1-1-2	1.33	low	Churned
8	PF-19120	835	4	9,062.864	1-1-5	2.33	high	Churned
9	DP-13165	812	2	1,058.616	1-1-2	1.33	low	Churned
10	CC-12685	799	4	2,854.115	1-2-4	2.33	high	Churned

```
--indi aldigimiz neticeni view ucun yaradaq:
CREATE OR REPLACE VIEW RFM_ANALYSIS AS
WITH max_table_date AS (
        SELECT
            max(order_date) AS max_date
```

```
        FROM RETAIL_SALES
),
RFM AS(
        SELECT
                customer_id,
                m.max_date - MAX(order_date)        AS recency,
                COUNT(DISTINCT(order_id))                   AS frequency,
                SUM(sales)                                          AS monetary
        FROM RETAIL_SALES r
        CROSS JOIN max_table_date m
GROUP BY r.customer_id, m.max_date
),
RFM_score AS(
        SELECT
                customer_id,
                recency,
                frequency,
                monetary,
                NTILE(5) OVER(ORDER BY recency DESC)   AS r_score,
                NTILE(5) OVER(ORDER BY frequency )              AS f_score,
                NTILE(5) over(ORDER BY monetary )               AS m_score
        FROM RFM
),
avg_rfm_concat AS(
        SELECT
                customer_id,
```

```
            recency,
            frequency,
            monetary,
            r_score,
            f_score,
            m_score,
            r_score ||'-'|| f_score ||'-'|| m_score AS R_F_M,
            ROUND((r_score + f_score + m_score) / 3, 2) AS AVG_rfm
        FROM rfm_score
),
value_seg AS(
        SELECT
            customer_id,
            recency,
            frequency,
            monetary,
            R_F_M,
            AVG_rfm,
            r_score,
            f_score,
            m_score,
            CASE
                WHEN m_score in(1,2) THEN 'low'
                WHEN m_score = 3    THEN 'mid'
                WHEN m_score in(4,5) THEN 'high'
            END AS value_segment
```

```
        FROM avg_rfm_concat
),
customer_seg as(
        SELECT
                customer_id,
                recency,
                frequency,
                monetary,
                R_F_M,
                AVG_RFM,
                value_segment,
                CASE
                        WHEN r_score >=4 AND f_score >=4 AND m_score >=4 THEN 'VIP'
                        WHEN f_score >=3 AND m_score <4 THEN 'Regular'
                        WHEN r_score <=3 AND r_score>1 THEN 'Dormat'
                        WHEN r_score =1 THEN 'Churned'
                        WHEN r_score >=4 AND f_score <=4 THEN 'New Customer'
                END AS cust_seg
        FROM value_seg
)
SELECT * FROM customer_seg


--view yaradildi.


--value segmentlerinde nece musterimiz olduguna baxaq:
SELECT
```

```
    value_segment,
    COUNT(customer_id) AS cust_count
FROM RFM_ANALYSIS
GROUP BY VALUE_SEGMENT
```

	VALUE_SEGMENT	CUST_COUNT
1	low	318
2	mid	159
3	high	316

--customer segmente gore musterilerimizin sayina baxaq:

```
SELECT
    CUST_SEG,
    COUNT(CUSTOMER_ID) AS cust_count
FROM RFM_ANALYSIS
GROUP BY CUST_SEG
```

	CUST_SEG	CUST_COUNT
1	Churned	133
2	Regular	215
3	Dormat	231
4	New Customer	118
5	VIP	96

--her segmentdeki musteriler umumi segmentin nece faizini teskil edir?

```sql
--Müştəri dəyərinə(monetary) görə Low / Mid / High Value yarat.
value_seg AS(
    SELECT
        customer_id,
        recency,
        frequency,
        monetary,
        R_F_M,
        AVG_rfm,
        r_score,
        f_score,
        m_score,
        CASE
            WHEN m_score in(1,2) THEN 'low'
            WHEN m_score = 3      THEN 'mid'
            WHEN m_score in(4,5) THEN 'high'
        END AS value_segment
    FROM avg_rfm_concat
),
```


```sql
--Müştəri dəyərinə(monetary) görə Low / Mid / High Value yarat.
value_seg AS(
    SELECT
        customer_id,
        recency,
        frequency,
        monetary,
        R_F_M,
        AVG_rfm,
        r_score,
        f_score,
        m_score,
        CASE
            WHEN m_score in(1,2) THEN 'low'
            WHEN m_score = 3      THEN 'mid'
            WHEN m_score in(4,5) THEN 'high'
        END AS value_segment
    FROM avg_rfm_concat
),
```

	CUST_SEG	CUST_COUNT
1	Churned	133
2	Regular	215
3	Dormat	231
4	New Customer	118
5	VIP	96